

Yuwen Cui

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RESEARCH INTERESTS

- Side-channel attack in system security: Analyzing system and hardware resource consumption patterns to identify privacy leakage during data processing and computation.
- Side-channel attack in cybersecurity: Analyzing network traffic patterns to infer users' activities during website access.
- Side-channel attack in AI security: Analyzing user-AI agent network traffic to infer user private information like career, identify AI models/versions, and recover prompt-related information.

EDUCATION

University of South Florida Tampa, FL	08/2021 – 12/2026
• Ph.D. in Computer Science and Engineering	
University of North Texas Denton, TX	01/2019 – 08/2021
• Graduate Coursework in Computer Science and Engineering	
China University of Petroleum Qingdao, China	08/2015 – 05/2018
• M.S. in Computer Science and Engineering	
Lanzhou University of Technology Lanzhou, China	09/2011 – 05/2015
• B.S. in Computer Science and Engineering	

Skills

- **Programming Languages:** Python, C/C++, Java, SQL, Bash
- **Machine Learning Tools:** PyTorch, TensorFlow, Scikit-learn, Hugging Face
- **Systems:** Linux, Git, Docker, Kubernetes
- **Research:** Experimental Design, Data Analysis, Privacy-Preserving ML, Network Security, AI Security

EXPERIENCE

University of South Florida Tampa, FL	08/2021 - Current
Research Assistant:	
• Conducted cybersecurity research by capturing and analyzing users' network traffic to infer their online activities and behaviors.	
• Conducted AI security research by analyzing user-AI agent network traffic to infer user private information like career, identify AI models/versions, and recover prompt-related information.	
• Conducted system security by analyzing system and hardware resource consumption patterns to identify privacy leakage during data processing and computation.	
• Designed, implemented, and evaluated time-series LLM prototypes for website fingerprinting attacks using Python, PyTorch, and Hugging Face.	

- Designed, implemented, and evaluated website fingerprinting defense prototype using Python, PyTorch, and Scikit-learn.
- Designed, implemented, and evaluated system and hardware resource consumption pattern recognition prototypes to infer users' co-located cloud services using C/C++ and bash.

Teaching Assistant:

- Led laboratory sessions and provided hands-on guidance with SQL injection, XSS, and other web security Labs to 100+ undergraduate students.
- Supported instruction in undergraduate courses on secure coding and Machine Learning, enhancing student understanding of key concepts.
- Led penetration testing sessions and provided hands-on guidance on *password cracking with Hashcat*, *web reconnaissance*, *the using of Metasploit Framework*, etc.

University of South Florida | Tampa, FL

05/2026 - 08/2026

NSF REU Mentor:

- Mentored undergraduate researchers in NSF REU cybersecurity project on website fingerprinting and encrypted traffic analysis, guiding students in applying deep learning techniques for traffic classification, evaluating fingerprint attacks and defenses, and conducting experiments to address privacy challenges in network security.
- Guided students through academic and personal challenges effectively.
- Facilitated workshops on study skills and time management techniques.

University of North Texas | Denton, TX

01/2019 - 07/2021

Research Assistant:

- Designed and implemented NoC architecture, evaluated system performance, and analyzed experimental data using state-of-the-art research methodologies.
- Developed research prototypes and reproducible experiments using Python, C/C++, Linux, and CUDA.
- Collaborated with faculty and graduate researchers to develop and submit projects to academic conferences, enhancing research visibility.

Teaching Assistant:

- Supported instruction in undergraduate courses in Computer Organization and Architecture course with 100+ students.
- Conducted ARM programming labs and supported student learning.
- Conducted weekly office hours to clarify course material and assist students with programming project debugging.
- Graded assignments, projects, and exams using standardized rubrics.

PROJECTS

Exploring the LLM User Privacy Leakage through Traffic Fingerprinting

08/2026 - current

Investigating privacy risks in encrypted traffic generated during interactions with LLM-based services and AI agents, and studying whether a passive traffic observer can analyze encrypted network traces to infer information: 1) User prompt characteristics — identifying or characterizing the type, category, or properties of prompts submitted to an LLM without directly observing their plaintext content. 2) LLM/agent identification - determining which LLM, AI agent, or service is being used based on its network traffic patterns. 3) Model/version fingerprinting - investigating whether different model or

agent versions produce distinguishable traffic signatures. 4) Interaction behavior - examining whether traffic patterns can reveal properties of multi-turn conversations, agentic workflows, tool usage, or different response behaviors.

Privacy-Preserving Defenses Against Website Fingerprinting Attacks

09/2025 - current

Developed privacy-preserving defense mechanisms against website fingerprinting attacks on encrypted traffic. Evaluated the robustness of existing defenses against adaptive machine learning-based attackers and explored techniques to improve privacy protection while maintaining network performance.

- *Github:* <https://github.com/WF-Attack-and-Defense/chameleon>

Time-Series Large Language Models for Website Fingerprinting Attacks

09/2025 - 02/2026

Developed a robust website fingerprinting framework that leverages time-series large language models (TSLMs) to improve resilience against model drift in encrypted traffic analysis. TSLM addresses the challenge of evolving network environments by learning long-term temporal traffic patterns and adapting to changes in website behaviors over time. The project explores the application of LLM-based sequence modeling for privacy analysis and demonstrates improved robustness against real-world traffic distribution shifts.

- *Github:* <https://github.com/WF-Attack-and-Defense/wf-tslm>

Wi-Fi Fingerprint-Based Device Localization and Tracking in Public Wi-Fi Networks

05/2024 - 05/2025

Developed a machine learning-based device localization and tracking system using Wi-Fi fingerprinting techniques with Received Signal Strength Indicator (RSSI) and Round-Trip Time (RTT) measurements. The project extracts wireless signal features from public Wi-Fi environments and applies machine learning models to estimate device locations and improve localization accuracy under dynamic network conditions. This work investigates the challenges of signal variability, wireless positioning, and privacy risks associated with device tracking in public Wi-Fi networks.

- *Github:* <https://github.com/RxStalker/RxStalker>

Shared Resource Vulnerability Analysis in Serverless Computing

05/2022 - 05/2023

Investigated security vulnerabilities in multi-tenant serverless computing environments by analyzing shared resource interference between co-resident serverless functions. Developed and evaluated a Shared Resource Entanglement (SRE) attack that demonstrates how attackers can infer sensitive information through memory access and resource usage patterns, highlighting potential data leakage risks in Kubernetes-based serverless platforms and motivating stronger isolation mechanisms.

PUBLICATIONS

Conference proceedings:

- **Yuwen Cui**, Guangjing Wang, Kai Wei, Kehan Shen, Yao Liu, Exploiting Wi-Fi Fingerprints for Practical Device Localization and Tracking in Public Environments, Accepted by *IEEE Annual Computer Security Applications Conference (ACSAC)* 2026.
- Kai Wei, **Yuwen Cui**, Kehan Shen, Guangjing Wang, A Multi-Agent Framework for High-Interaction Terminal Simulation, Accepted by *The 64th Annual Meeting of the Association for Computational Linguistics (ACL)* 2026.
- **Yuwen Cui**, Mingkun Wei, Zhuo Lu, Yao Liu, Shared Resource Entanglement Attacks against Serverless Computing, Accepted by *2024 IEEE Conference on Communications and Network Security (CNS)*.
- **Yuwen Cui**, et. al., A Low-Cost Conflict-Free NoC Architecture for Heterogeneous Multicore Systems, Accepted by *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)* 2020.

Journal articles:

- L Shi, **Y Cui**, X Han, H Chen, D Liu, Mimicry honeypot: an evolutionary decoy system, Accepted by *International Journal of High Performance Computing and Networking* 2019.

Preprints:

- **Yuwen Cui**, et al., A comprehensive survey of website fingerprinting attacks and defenses in tor: Advances and open challenges, arXiv preprint 2025.
- **Yuwen Cui**, et al., Chameleon: Robust Defense Against Tor Website Fingerprinting via Many-to-Many Traffic Morphing, arXiv preprint 2026.

Posters:

- **Yuwen Cui**, Kai Wei, Kehan Shen, Guangjing Wang, Poster: Towards Model Drift Resistant Website Fingerprinting with Time-Series LLMs, Accepted by *Network and Distributed System Security (NDSS) Symposium* 2026.

TEACHING EXPERIENCE

Graduate Courses Teaching Assistant:

- CIS6220 - Penetration Testing for IT, University of South Florida, Fall 2026.
- CNT6410 - Emerging Topics in Network Security, University of South Florida, Spring 2025.

Undergraduate Courses Teaching Assistant:

- CIS4200 - Penetration Testing for IT, University of South Florida, Fall 2026.
- CAI5107 - Machine Learning, University of South Florida, Fall 2026.
- CNT4419 - Secure Coding, University of South Florida, Spring 2022, Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Fall 2025, Spring 2026.
- CNT6410 - Emerging Topics in Network Security, University of South Florida, Spring 2025.
- CSCE 2610 - Assembly Language and Computer Organization, University of North Texas, Fall 2019, Spring 2020, Summer 2020, Fall 2020, Spring 2021, Summer 2021.